

Umberto De Luca (he/him) Computer Engineering

Kelowna, British Columbia

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TECHNICAL SKILLS

- **Programming Languages:** Lua, Python, C#, Java, C, MATLAB
- **Frameworks/Libraries:** FastAPI, SQLAlchemy, Pydantic, TensorFlow, Scikit-learn, NLTK, JUnit, LOVE2D
- **Development Tools:** Git/GitHub, Cursor/Codex, Unity, Visual Studio Code, Visual Studio, SolidWorks, Microsoft Office, Make

EDUCATION

University of British Columbia

2022 - Present

Bachelor of Applied Science - Computer Engineering

CGPA: 80.0%

Related Courses:

- *Computer Programming I:* 99%
- *Computer Fluency:* 91%
- *Digital Logic Design:* 90%

Pontificia Universidade Católica do Rio de Janeiro

2020 - 2022

Mechanical Engineering

CGPA: 86%

TECHNICAL PROJECTS

Food Delivery Backend API, Course Project

February 2026 – April 2026

- Designed and implemented a modular backend using MVC architecture, separating model, routing, controller, repository, and service layer logic using Python, VS Code and Git/GitHub.
- Developed the order and delivery management system, including order creation, modification, and state transitions with robust validation using Pydantic.
- Built a payment processing system combining a custom simulated transaction engine with the PayPal Sandbox API, handling transaction validation and edge cases to support realistic payment workflows and testing.
- Wrote unit and integration tests using Pytest, applying equivalence partitioning and mock-based testing for reliability.

Snake Game, Personal Project

June 2024 - July 2024

- Developed a fully functional code and interface for a "Snake Game" with state machines for each game scene - start, play, and game over – using Lua, Visual Studio Code, and Love2d features
- Leveraged Lua's Knife library and its Timer.tween method to deliver an engaging gameplay experience by simulating a snake's movement across the board grid
- Built a modified A* pathfinding algorithm for a Snake AI, enabling fully autonomous gameplay and ensuring consistent survival in the "start" state

OTHER WORK EXPERIENCE



THE UNIVERSITY OF BRITISH COLUMBIA

Co-op Program
Faculty of Applied Science

COOP.APSC.UBC.CA
604-822-3022
apsc.coop@ubc.ca

Totaltag Soluções em Informática Ltda, Rio de Janeiro, Brazil

June 2026 – September 2026

Technical Intern

- Mapped and documented the architecture and feature flows of a full-stack web application to support development and maintenance.
- Developed and enhanced features for a dental clinic management platform using Python, FastAPI, Pydantic, SQLAlchemy with SQLite, React, and Pytest.
- Leveraged AI-assisted development tools (Cursor, Codex, and large language models) to accelerate feature development, debugging, refactoring, and code analysis while validating generated code and maintaining software quality.

University of British Columbia Food Services, Kelowna, Canada

September 2024 – April 2026

Third Cook (Auxiliary)

- Collaborated with the cleaning staff to provide a safe and clean environment for customers
- Worked in a fast-paced environment, assisting the first and second cooks during their routines

Datasphere Dynamics, Vancouver, Canada

May 2025 – September 2025

Intern

- Earned Microsoft M365 Fundamentals and Azure Fundamentals certifications as part of cloud training
- Supported customers in recovering compromised accounts and strengthening security posture, including endpoint hardening and account protection

Pontifícia Universidade Católica, Rio de Janeiro, Brazil

March 2021 - December 2021

Linear Algebra & Calculus I Tutor

- Supported students who were struggling with Linear Algebra and Calculus I by solving and giving visual examples using MATLAB
- Collaborated with the course Professor to supervise students during examinations

ENGINEERING OR DESIGN STUDENT TEAMS

Reptiles-Baja, Pontifícia Universidade Católica do Rio de Janeiro

August 2021 - April 2022

Suspension & Steering Team Member

- Collaborated with other members to build and design the suspension and steering system of the Baja Car using SolidWorks
- Performed practical and theoretical tests using MATLAB to understand and adjust the steering and suspension settings of the vehicle

VOLUNTEER EXPERIENCE

Pontifícia Universidade Católica, Rio de Janeiro, Brazil

March 2022 - July 2022

Mathematics Assistant Professor at the “Seja Mais” Project

- Tutored a class of over 15 undergraduate mature students for university admissions exams
- Collaborated with the Mathematics Project Coordinator to provide an adequate curriculum

INTERESTS & ACTIVITIES

- **Completed the Introduction to Game Development and Introduction to AI free course version from Harvard edX - CS50**
- **Won the 2017 state and South American basketball U15 championship with “Clube de Regatas do Flamengo”**



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